

Chapter 8

eFuse Controller (EFUSE)

8.1 Overview

ESP32-P4 contains a 4096-bit eFuse memory to store parameters and user data. The parameters include control parameters for some hardware modules, system data parameters and keys used for the encryption/decryption module. Once an eFuse bit is programmed to 1, it can never be reverted to 0. The eFuse controller programs individual bits of parameters in eFuse according to user configurations. From outside the chip, eFuse data can only be read via the eFuse controller. For some data, such as some keys stored in eFuse for internal use by hardware cryptography modules (e.g., digital signature, HMAC), if read protection is not enabled, the data can be read from outside the chip; if read protection is enabled, the data cannot be read from outside the chip.

8.2 Features

- 4096-bit one-time programmable memory (including up to 1792 bits reserved for custom use depending on whether the EFUSE_KEY_PURPOSE_*n* parameters are set to 0. For more information, see Table 8.3-2.)
- Configurable write protection
- Configurable read protection
- Various hardware encoding schemes against data corruption

8.3 Functional Description

8.3.1 Structure

The eFuse system consists of the eFuse controller and eFuse memory. Data flow in this system is shown in Figure 8.3-1.

Users can program bits in the eFuse memory via the eFuse controller by writing the data to be programmed to the programming register and executing the programming instruction. For detailed programming steps, please refer to Section 8.3.2.

Users cannot directly read the data programmed in the eFuse memory, so they need to read the programmed data into the Reading Data Register of the corresponding address segment through the eFuse controller. During the reading process, if the data is inconsistent with that in the eFuse memory, the eFuse controller can automatically correct it through the hardware encoding mechanism (see Section 8.3.1.3 for details), and send

the error message to the error report register. For detailed steps to read parameters, please refer to the Section [8.3.3](#).

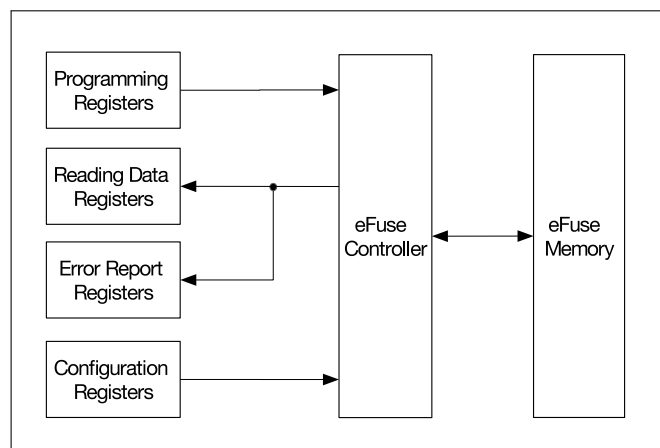


Figure 8.3-1. Data Flow in eFuse

Data in eFuse memory is organized in 11 blocks (BLOCK0 ~ BLOCK10).

BLOCK0 holds most parameters for software and hardware uses.

Table [8.3-1](#) lists all the parameters accessible (readable and usable) to users in BLOCK0 and their offsets, bit widths, accessibility by hardware, write protection, and brief function description. For more description on the parameters, please click the link of the corresponding parameter in the table.

The [EFUSE_WR_DIS](#) parameter is used to disable write protection of other parameters. [EFUSE_RD_DIS](#) is used to disable read protection of BLOCK4 ~ BLOCK10. For more information on these two parameters, please see Section [8.3.1.1](#) and Section [8.3.1.2](#).

Table 8.3-1. Parameters in eFuse BLOCK0

Parameters	Bit Width	Accessible by Hardware	Write Protection by EFUSE_WR_DIS Bit Number	Description
EFUSE_WR_DIS	32	Y	N/A	Represents whether writing of eFuse bits by eFuse controller is disabled.
EFUSE_RD_DIS	7	Y	0	Represents whether reading data from BLOCK4 ~ 10 in eFuse memory by users is disabled.
EFUSE_USB_DEVICE_EXCHG_PINS	1	Y	30	Represents whether the USB Serial/JTAG pins D+ and D- are exchanged.
EFUSE_USB_OTG11_EXCHG_PINS	1	Y	30	Represents whether the USB FS pins D+ and D- are exchanged.
EFUSE_DIS_USB_JTAG	1	Y	2	Represents whether the USB-to-JTAG function in USB Serial/JTAG module is disabled.
EFUSE_POWERGLITCH_EN	1	Y	2	Represents whether power glitch detection is enabled.
EFUSE_DIS_FORCE_DOWNLOAD	1	Y	2	Represents whether the function to force the chip into Download mode is disabled.
EFUSE_SPI_DOWNLOAD_MSPI_DIS	1	Y	2	Represents accessing MSPI flash/MSPI RAM by SYS AXI matrix is disabled during boot_mode_download.
EFUSE_DIS_TWAI	1	Y	2	Represents whether the TWAI controller is disabled.
EFUSE_JTAG_SEL_ENABLE	1	Y	2	Represents whether the selection of a JTAG signal source through the strapping value is enabled when both EFUSE_DIS_PAD_JTAG and EFUSE_DIS_USB_JTAG are configured to 0.
EFUSE_SOFT_DIS_JTAG	3	Y	31	Represents whether PAD JTAG is disabled in the soft way.
EFUSE_DIS_PAD_JTAG	1	Y	2	Represents whether PAD JTAG is disabled in the hard way (permanently).
EFUSE_DIS_DOWNLOAD_MANUAL_ENCRYPT	1	Y	2	Represents whether flash encryption is disabled (except in SPI boot mode).

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Table 8.3-1 – cont'd from previous page

Parameters	Bit Width	Accessible by Hardware	Write Protection by EFUSE_WR_DIS Bit Number	Description
EFUSE_USB_PHY_SEL	1	Y	30	Represents the connection between USB Serial/JTAG Controller, OTG_FS and FS_PHY1, FS_PHY2.
EFUSE_XTS_KEY_LENGTH_256	1	N	1	Represents which key is used for flash encryption.
EFUSE_WDT_DELAY_SEL	2	Y	2	Represents RTC watchdog timeout threshold.
EFUSE_SPI_BOOT_CRYPT_CNT	3	Y	4	Represents whether SPI boot encryption/decryption is enabled.
EFUSE_SECURE_BOOT_KEY_REVOKE0	1	N	5	Represents whether revoking the first Secure Boot key is enabled.
EFUSE_SECURE_BOOT_KEY_REVOKE1	1	N	6	Represents whether revoking the second Secure Boot key is enabled.
EFUSE_SECURE_BOOT_KEY_REVOKE2	1	N	7	Represents whether revoking the third Secure Boot key is enabled.
EFUSE_KEY_PURPOSE_0	4	Y	8	Represents Key0 purpose. See Table 8.3-2.
EFUSE_KEY_PURPOSE_1	4	Y	9	Represents Key1 purpose. See Table 8.3-2.
EFUSE_KEY_PURPOSE_2	4	Y	10	Represents Key2 purpose. See Table 8.3-2.
EFUSE_KEY_PURPOSE_3	4	Y	11	Represents Key3 purpose. See Table 8.3-2.
EFUSE_KEY_PURPOSE_4	4	Y	12	Represents Key4 purpose. See Table 8.3-2.
EFUSE_KEY_PURPOSE_5	4	Y	13	Represents Key5 purpose. See Table 8.3-2.
EFUSE_SEC_DPA_LEVEL	2	Y	14	Represents the security level of anti-DPA (differential power analysis) attack.
EFUSE_CRYPT_DPA_ENABLE	1	Y	14	Represents whether defense against DPA attack is enabled.
EFUSE_SECURE_BOOT_EN	1	N	15	Represents whether Secure Boot is enabled.
EFUSE_SECURE_BOOT_AGGRESSIVE_REVOKE	1	N	16	Represents whether aggressive revocation of Secure Boot is enabled.
EFUSE_FLASH_TYPE	1	N	18	Represents the type of the interfaced flash.

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Table 8.3-1 – cont'd from previous page

Parameters	Bit Width	Accessible by Hardware	Write Protection by EFUSE_WR_DIS Bit Number	Description
EFUSE_FLASH_PAGE_SIZE	2	N	18	Represents flash page size.
EFUSE_FLASH_ECC_EN	1	N	18	Represents whether ECC is enabled during flash boot.
EFUSE_DIS_USB_OTG_DOWNLOAD_MODE	1	N	18	Represents whether download via USB-OTG is disabled.
EFUSE_FLASH_TPUW	4	N	18	Represents the flash waiting time after power-up.
EFUSE_DIS_DOWNLOAD_MODE	1	N	18	Represents whether all Download modes are disabled.
EFUSE_DIS_DIRECT_BOOT	1	N	18	Represents whether direct boot mode is disabled.
EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT	1	N	18	Represents whether print from USB-Serial-JTAG during ROM boot is disabled.
EFUSE_DIS_USB_SERIAL_JTAG_DOWNLOAD_MODE	1	N	18	Represents whether the USB-Serial-JTAG download function is disabled.
EFUSE_ENABLE_SECURITY_DOWNLOAD	1	N	18	Represents whether security download is enabled.
EFUSE_UART_PRINT_CONTROL	2	N	18	Represents the type of UART printing.
EFUSE_FORCE_SEND_RESUME	1	N	18	Represents whether ROM code is forced to send a resume command during SPI boot.
EFUSE_SECURE_VERSION	16	N	18	Represents the version used by ESP-IDF anti-rollback feature.
EFUSE_SECURE_BOOT_DISABLE_FAST_WAKE	1	N	19	Represents whether FAST VERIFY ON WAKE is disabled or enabled when Secure Boot is enabled.
EFUSE_HYS_EN_PAD	1	Y	2	Represents whether the hysteresis function of PAD0 – PAD27 is enabled.
EFUSE_DCDC_VSET	5	Y	N/A	Represents the default DCDC voltage.
EFUSE_OPXA_TIEH_SELO	2	Y	2	Represents which output source is used for Extern Power (V01).
EFUSE_OPXA_TIEH_SEL1	2	Y	2	Represents which output source is used for Extern Power (V02).

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Table 8.3-1 – cont'd from previous page

Parameters	Bit Width	Accessible by Hardware	Write Protection by EFUSE_WR_DIS Bit Number	Description
EFUSE_OPXA_TIEH_SEL2	2	Y	2	Represents which output source is used for Extern Power (VO3).
EFUSE_OPXA_TIEH_SEL3	2	Y	2	Represents which output source is used for Extern Power (VO4).
EFUSE_HP_PWR_SRC_SEL	1	Y	3	Represents the HP system power source.
EFUSE_DCDC_VSET_EN	1	Y	N/A	Represents whether to use the default voltage configured by EFUSE_DCDC_VSET .
EFUSE_DIS_WDT	1	Y	2	Represents whether to disable the watchdog.
EFUSE_DIS_SWD	1	Y	2	Represents whether to disable the super watchdog.

Table 8.3-2 lists all key purposes and their values. Setting the eFuse parameter EFUSE_KEY_PURPOSE_*n* declares the purpose of KEY*n* (*n*: 0 ~ 5).

Table 8.3-2. Secure Key Purpose Values

Key Purpose Values	Purposes
0	User purposes
1	ECDSA_KEY
2	XTS_AES_256_KEY_1
3	XTS_AES_256_KEY_2
4	XTS_AES_128_KEY (flash/SRAM encryption and decryption)
5	HMAC Downstream mode (both JTAG and RSA_DS)
6	JTAG in HMAC Downstream mode
7	RSA_DS peripheral in HMAC Downstream mode
8	HMAC Upstream mode
9	SECURE_BOOT_DIGEST0 (secure boot key digest)
10	SECURE_BOOT_DIGEST1 (secure boot key digest)
11	SECURE_BOOT_DIGEST2 (secure boot key digest)

Table 8.3-3 provides the details of parameters in BLOCK1 ~ BLOCK10.

Table 8.3-3. Parameters in BLOCK1 to BLOCK10

BLOCK	Parameters	Bit Width	Accessible by Hardware	Write Protection by EFUSE_WR_DIS Bit Number	Read Protection by EFUSE_RD_DIS Bit Number	Description
BLOCK1	EFUSE_MAC	48	N	20	N/A	MAC address
	EFUSE_MAC_EXT	16	N	20	N/A	Extended MAC address
	EFUSE_MAC_RESERVED	50	N	20	N/A	Reserved MAC address
	EFUSE_SYS_DATA_PART0	78	N	20	N/A	System data
BLOCK2	EFUSE_SYS_DATA_PART1	256	N	21	N/A	System data
BLOCK3	EFUSE_USR_DATA	256	N	22	N/A	User data
BLOCK4	EFUSE_KEY0_DATA	256	Y	23	0	KEY0 or user data
BLOCK5	EFUSE_KEY1_DATA	256	Y	24	1	KEY1 or user data
BLOCK6	EFUSE_KEY2_DATA	256	Y	25	2	KEY2 or user data
BLOCK7	EFUSE_KEY3_DATA	256	Y	26	3	KEY3 or user data
BLOCK8	EFUSE_KEY4_DATA	256	Y	27	4	KEY4 or user data
BLOCK9	EFUSE_KEY5_DATA	256	Y	28	5	KEY5 or user data
BLOCK10	EFUSE_SYS_DATA_PART2	256	N	29	6	System data

Among these blocks, BLOCK4 ~ 9 can be used to store KEY0 ~ 5. Up to six 256-bit keys can be written into eFuse. Whenever a key is written, its purpose value should also be written (see table 8.3-2). For example, when a key for the JTAG function in HMAC Downstream mode is written to KEY3 (i.e., BLOCK7), its key purpose value 6 should also be written to EFUSE_KEY_PURPOSE_3.

Note:

Do not program the XTS-AES key into the KEY5 block, i.e., BLOCK9. Otherwise, the key may be unreadable. Instead, program it into the preceding blocks, i.e., BLOCK4 ~ BLOCK8. The last block, BLOCK9, is used to program other keys.

BLOCK1 ~ BLOCK10 use the RS coding scheme, so there are some limitations on writing to these parameters. For more detailed information, please refer to Section 8.3.1.3 and Section 8.3.2.

8.3.1.1 EFUSE_WR_DIS

Parameter EFUSE_WR_DIS determines whether individual eFuse parameters are write-protected. After EFUSE_WR_DIS has been programmed, execute an eFuse read operation so the new values would take effect.

Column “Write Protection by EFUSE_WR_DIS Bit Number” in Table 8.3-1 and Table 8.3-3 list the specific bits in EFUSE_WR_DIS that disable writing.

When the write protection bit of a parameter is set to 0, it means that this parameter is not write-protected and can be programmed, unless it has been programmed before.

When the write protection bit of a parameter is set to 1, it means that this parameter is write-protected and none of its bits can be modified, with non-programmed bits always remaining 0 and programmed bits always remaining 1. That is to say, if a parameter is write-protected, it will always remain in this state and cannot be changed.

8.3.1.2 EFUSE_RD_DIS

Only the parameters in BLOCK4 ~ BLOCK10 can be set to be read-protected from users, as shown in column “Read Protection by EFUSE_RD_DIS Bit Number” of Table 8.3-3. After EFUSE_RD_DIS has been programmed, execute an eFuse read operation so the new values would take effect.

If the corresponding EFUSE_RD_DIS bit is 0, the parameter controlled by this bit is not read-protected from users. If it is 1, the parameter controlled by it is read-protected from users.

Other parameters that are not in BLOCK4 ~ BLOCK10 can always be read by users.

When BLOCK4 ~ BLOCK10 are set to be read-protected, the data in them can still be read by hardware cryptography modules if the EFUSE_KEY_PURPOSE_ *n* bit is set accordingly.

8.3.1.3 Data Storage

Internally, eFuse uses the hardware encoding scheme to protect data from corruption. The scheme and the encoding process are invisible to users.

All BLOCK0 parameters except for EFUSE_WR_DIS are stored with four backups, meaning each bit is stored four times. This backup scheme is not visible to users.

In BLOCK0, [EFUSE_WR_DIS](#) occupies 32 bits, and other parameters takes 152 bits each. So, the eFuse memory space occupied by BLOCK0 is $32 + 152 * 4 = 640$ bits.

BLOCK1 ~ BLOCK10 use RS (44, 32) coding scheme that supports up to 6 bytes of automatic error correction. The primitive polynomial of RS (44, 32) is $p(x) = x^8 + x^4 + x^3 + x^2 + 1$.

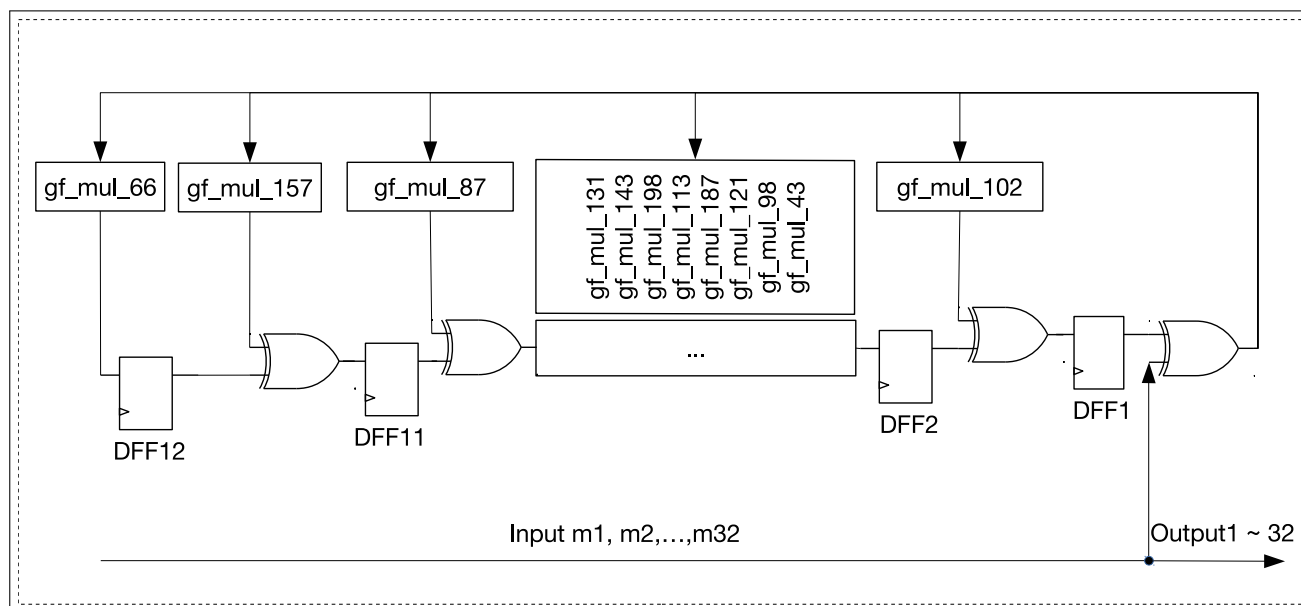


Figure 8.3-2. Shift Register Circuit (first 32 output)

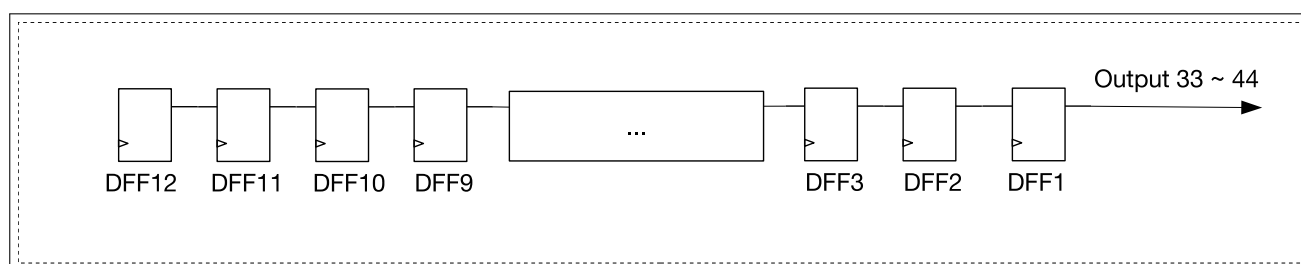


Figure 8.3-3. Shift Register Circuit (last 12 output)

The shift register circuit shown in Figure 8.3-2 and 8.3-3 processes 32 data bytes using RS (44, 32). This coding scheme encodes 32 bytes of data into 44 bytes:

- Bytes [0:31] are the data bytes itself
- Bytes [32:43] are the encoded parity bytes stored in 8-bit flip-flops DFF1, DFF2, ..., DFF12 (gf_mul_n is the result of multiplying a byte of data in $GF(2^8)$ by α^n , where n is an integer).

After that, the hardware programs into eFuse the 44-byte codeword consisting of the data bytes and the parity bytes. When the eFuse block is read, the eFuse controller automatically decodes the codeword and applies error correction if needed.

Because the RS check codes are generated on the entire 32-byte eFuse block, each block can only be written once.

Since the size of BLOCK1 is less than 32 bytes, the unused bits will be treated as 0 by hardware during the RS (44, 32) encoding. Thus, the final coding result will not be affected.

Among blocks using the RS (44, 32) coding scheme, the parameters in BLOCK1 is 24 bytes, and the RS check code is 12 bytes, so BLOCK1 occupies $24 + 12 = 36$ bytes in eFuse memory.

The parameter in other blocks (Block2 ~ 10) is 32 bytes respectively, and the RS check code is 12 bytes, so they occupy $(32 + 12) * 9 = 396$ bytes in eFuse memory.

8.3.2 Programming of Parameters

The eFuse controller can only program eFuse parameters in one block at a time. BLOCK0 ~ BLOCK10 share the same address range to store the parameters to be programmed. Configure parameter [EFUSE_BLK_NUM](#) to indicate which block should be programmed.

Since there is a one-to-one correspondence between the reading data registers and the programming data registers (see table 8.3-4 for details), users can find out where the data to be programmed is located in programming registers by checking the parameter description and the parameter location in the corresponding read registers.

For example, if the user wants to program the parameter [EFUSE_USB_DEVICE_EXCHG_PINS](#) in BLOCK0 to 1, they can first search the reading data registers [EFUSE_RD_REPEAT_DATA0 ~ 4_REG](#) in BLOCK0 for where the parameter is located, namely, the 8th bit in [EFUSE_RD_REPEAT_DATA0_REG](#). So, the user can set the 8th bit of [EFUSE_PGM_DATA1_REG](#) to 1 and follow the programming steps below. After the steps are completed, the corresponding bit in the eFuse memory will be programmed to 1.

Programming preparation

- Programming BLOCK0
 1. Set [EFUSE_BLK_NUM](#) to 0.
 2. Write into [EFUSE_PGM_DATA0_REG ~ EFUSE_PGM_DATA5_REG](#) the data to be programmed to BLOCK0.
The data in [EFUSE_PGM_DATA6_REG ~ EFUSE_PGM_DATA7_REG](#) and [EFUSE_PGM_CHECK_VALUE0_REG ~ EFUSE_PGM_CHECK_VALUE2_REG](#) does not affect the programming of BLOCK0.
- BLOCK1 cannot be programmed by users as it has been programmed at manufacturing.
- Programming BLOCK2 ~ 10
 1. Set [EFUSE_BLK_NUM](#) to the block number.
 2. Write into [EFUSE_PGM_DATA0_REG ~ EFUSE_PGM_DATA7_REG](#) the data to be programmed. Write into [EFUSE_PGM_CHECK_VALUE0_REG ~ EFUSE_PGM_CHECK_VALUE2_REG](#) the corresponding RS code.

Programming process

The process of programming parameters is as follows:

1. Configure the value of parameter [EFUSE_BLK_NUM](#) to determine the block to be programmed.
2. Write parameters to be programmed to registers [EFUSE_PGM_DATA0_REG ~ EFUSE_PGM_DATA7_REG](#) and [EFUSE_PGM_CHECK_VALUE0_REG ~ EFUSE_PGM_CHECK_VALUE2_REG](#).
3. Make sure the eFuse programming voltage VDDQ is configured correctly as described in Section 8.3.4.
4. Configure the field [EFUSE_OP_CODE](#) of register [EFUSE_CONF_REG](#) to 0x5A5A.

5. Configure the field [EFUSE_PGM_CMD](#) of register [EFUSE_CMD_REG](#) to 1.
6. Poll register [EFUSE_CMD_REG](#) until it is 0x0, or wait for a PGM_DONE interrupt. For more information on how to identify a PGM_DONE or READ_DONE interrupt, please see the end of Section 8.3.3.
7. Clear the parameters in [EFUSE_PGM_DATA0_REG](#) ~ [EFUSE_PGM_DATA7_REG](#) and [EFUSE_PGM_CHECK_VALUE0_REG](#) ~ [EFUSE_PGM_CHECK_VALUE2_REG](#).
8. Trigger an eFuse read operation (see Section 8.3.3) to update eFuse registers with the new values.
9. Check error record registers. If the values read in error record registers are not 0, the programming process should be performed again following above steps 1 ~ 7. Please check the following error record registers for different eFuse blocks:
 - BLOCK0: [EFUSE_RD_REPEAT_ERR0_REG](#) ~ [EFUSE_RD_REPEAT_ERR4_REG](#)
 - BLOCK1: [EFUSE_MAC_SYS_ERR_NUM](#), [EFUSE_MAC_SYS_FAIL](#)
 - BLOCK2: [EFUSE_SYS_PART1_ERR_NUM](#), [EFUSE_SYS_PART1_FAIL](#)
 - BLOCK3: [EFUSE_USR_DATA_ERR_NUM](#), [EFUSE_USR_DATA_FAIL](#)
 - BLOCK4: [EFUSE_KEY0_ERR_NUM](#), [EFUSE_KEY0_FAIL](#)
 - BLOCK5: [EFUSE_KEY1_ERR_NUM](#), [EFUSE_KEY1_FAIL](#)
 - BLOCK6: [EFUSE_KEY2_ERR_NUM](#), [EFUSE_KEY2_FAIL](#)
 - BLOCK7: [EFUSE_KEY3_ERR_NUM](#), [EFUSE_KEY3_FAIL](#)
 - BLOCK8: [EFUSE_KEY4_ERR_NUM](#), [EFUSE_KEY4_FAIL](#)
 - BLOCK9: [EFUSE_KEY5_ERR_NUM](#), [EFUSE_KEY5_FAIL](#)
 - BLOCK10: [EFUSE_SYS_PART2_ERR_NUM](#), [EFUSE_SYS_PART2_FAIL](#)

Limitations

In BLOCK0, each bit can be programmed separately. However, we recommend to minimize programming cycles and program all the bits of a parameter in one programming action. In addition, after all parameters controlled by a certain bit of [EFUSE_WR_DIS](#) are programmed, that bit should be immediately programmed. The programming of parameters controlled by a certain bit of [EFUSE_WR_DIS](#), and the programming of the bit itself can even be completed at the same time in one programming action.

BLOCK1 cannot be programmed by users as it has been programmed at manufacturing.

BLOCK2 ~ 10 can only be programmed once. Repeated programming is not allowed.

8.3.3 Reading of Parameters by Users

Users cannot read eFuse bits directly. The eFuse controller hardware reads all eFuse bits and stores the results to their corresponding registers in its memory space. Then, users can read eFuse bits by reading the registers that start with [EFUSE_RD_](#). Details are provided in Table 8.3-4.

Table 8.3-4. Registers Information

BLOCK	Read Registers	Registers When Programming This Block
0	EFUSE_RD_WR_DIS_REG	EFUSE_PGM_DATA0_REG
0	EFUSE_RD_REPEAT_DATA0 ~ 4_REG	EFUSE_PGM_DATA1 ~ 5_REG
1	EFUSE_RD_MAC_SYS_0 ~ 5_REG	EFUSE_PGM_DATA0 ~ 5_REG
2	EFUSE_RD_SYS_PART1_DATA0 ~ 7_REG	EFUSE_PGM_DATA0 ~ 7_REG
3	EFUSE_RD_USR_DATA0 ~ 7_REG	EFUSE_PGM_DATA0 ~ 7_REG
4-9	EFUSE_RD_KEY_n_DATA0 ~ 7_REG (<i>n</i> : 0 ~ 5)	EFUSE_PGM_DATA0 ~ 7_REG
10	EFUSE_RD_SYS_PART2_DATA0 ~ 7_REG	EFUSE_PGM_DATA0 ~ 7_REG

Updating reading data registers

The eFuse controller reads eFuse memory to update corresponding registers. This read operation happens at system reset and can also be triggered manually by users as needed (e.g., if new eFuse values have been programmed). The process of triggering a read operation by users is as follows:

1. Configure the field [EFUSE_OP_CODE](#) in register [EFUSE_CONF_REG](#) to 0x5AA5.
2. Configure the field [EFUSE_READ_CMD](#) in register [EFUSE_CMD_REG](#) to 1.
3. Poll register [EFUSE_CMD_REG](#) until it is 0x0, or wait for a READ_DONE interrupt. Information on how to identify a PGM_DONE or READ_DONE interrupt is provided below in this section.
4. Read the values of each parameter from eFuse memory.

The eFuse read registers will hold all values until the next read operation.

Error detection

The programming error record registers allows users to check the integrity of parameters stored in the eFuse memory. For instance, they can help detect whether the four-backup parameters are consistent and whether parameters protected by RS encoding are decoded successfully.

Registers [EFUSE_RD_REPEAT_ERRO ~ 3_REG](#) indicate if there are any errors in programming parameters (except [EFUSE_WR_DIS](#)) to BLOCK0. The value 1 indicates an error is detected in programming the corresponding bit. The value 0 indicates no error.

Registers [EFUSE_RD_RS_ERRO ~ 1_REG](#) store the number of corrected bytes as well as the result of RS decoding when eFuse controller reads BLOCK1 ~ BLOCK10.

The values of the above registers will be updated every time the reading data registers of eFuse controller have been updated.

Identifying completion of program or read operations

The methods to identify the completion of a program/read operation are described below. Please note that bit 1 corresponds to a program operation, and bit 0 corresponds to a read operation.

- Method one: Poll bit 1/0 in register [EFUSE_INT_RAW_REG](#) until it becomes 1, which represents the completion of a program/read operation.
- Method two:
 1. Set bit 1/0 in register [EFUSE_INT_ENA_REG](#) to 1 to enable the eFuse controller to post a PGM_DONE or READ_DONE interrupt.

2. Configure the Interrupt Matrix to enable the CPU to respond to eFuse interrupt signals. See Chapter [12 Interrupt Matrix](#).
3. Wait for the PGM_DONE or READ_DONE interrupt.
4. Set bit 1/0 in register [EFUSE_INT_CLR_REG](#) to 1 to clear the PGM_DONE or READ_DONE interrupt.

Note

When eFuse controller is updating its registers, it will use [EFUSE_PGM_DATA_n_REG](#) (n=0, 1, ... ,7) again to store data. So please do not write important data into these registers before this updating process is initiated.

During the chip boot process, eFuse controller will automatically update data from eFuse memory into the registers that can be accessed by users. Users can get programmed eFuse data by reading corresponding registers. Thus, there is no need to update the reading data registers in such case.

8.3.4 eFuse VDDQ Timing

The eFuse controller operates at the clock frequency of 20 MHz, and its programming voltage VDDQ should be configured as follows:

- [EFUSE_DAC_NUM](#) (the rising period of VDDQ): The default value of VDDQ is 2.5 V and the voltage increases by 0.01 V in each clock cycle. The default value of this parameter is 255.
- [EFUSE_DAC_CLK_DIV](#) (the clock divisor of VDDQ): The clock period to program VDDQ should be larger than 1 μ s.
- [EFUSE_PWR_ON_NUM](#) (the power-up time for VDDQ): The programming voltage should be stabilized after this time, which means the value of this parameter should be configured to exceed the result of [EFUSE_DAC_CLK_DIV](#) times [EFUSE_DAC_NUM](#).
- [EFUSE_PWR_OFF_NUM](#) (the power-out time for VDDQ): The value of this parameter should be larger than 10 μ s.

Table 8.3-5. Configuration of Default VDDQ Timing Parameters

EFUSE_DAC_NUM	EFUSE_DAC_CLK_DIV	EFUSE_PWR_ON_NUM	EFUSE_PWR_OFF_NUM
0xFF	0x28	0x3000	0x190

8.3.5 Parameters Used by Hardware Modules

Some hardware modules are directly connected to the eFuse peripheral in order to use the parameters that are marked with “Y” in columns “Accessible by Hardware” of Table [8.3-1](#) and Table [8.3-3](#). Users cannot intervene in this process.

8.4 Interrupts

ESP32-P4's eFuse controller can generate the following interrupt signal that will be sent to the [Interrupt Matrix](#).

- EFUSE_INT

There are several internal interrupt sources from eFuse controller that can generate the above interrupt signal. The interrupt sources from eFuse controller are listed with their trigger conditions and the resulted interrupt signal in Table 8.4-1.

Table 8.4-1. eFuse's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
EFUSE_PGM_DONE_INT	Programming of eFuse completes	EFUSE_INT
EFUSE_READ_DONE_INT	Reading of eFuse completes	

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [8.5 Register Summary](#).

8.5 Register Summary

The addresses in this section are relative to eFuse controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Programming Data Registers			
EFUSE_PGM_DATA0_REG	Register 0 that stores data to be programmed	0x0000	R/W
EFUSE_PGM_DATA1_REG	Register 1 that stores data to be programmed	0x0004	R/W
EFUSE_PGM_DATA2_REG	Register 2 that stores data to be programmed	0x0008	R/W
EFUSE_PGM_DATA3_REG	Register 3 that stores data to be programmed	0x000C	R/W
EFUSE_PGM_DATA4_REG	Register 4 that stores data to be programmed	0x0010	R/W
EFUSE_PGM_DATA5_REG	Register 5 that stores data to be programmed	0x0014	R/W
EFUSE_PGM_DATA6_REG	Register 6 that stores data to be programmed	0x0018	R/W
EFUSE_PGM_DATA7_REG	Register 7 that stores data to be programmed	0x001C	R/W
EFUSE_PGM_CHECK_VALUE0_REG	Register 0 that stores the RS code to be programmed	0x0020	R/W
EFUSE_PGM_CHECK_VALUE1_REG	Register 1 that stores the RS code to be programmed	0x0024	R/W
EFUSE_PGM_CHECK_VALUE2_REG	Register 2 that stores the RS code to be programmed	0x0028	R/W
Reading Data Registers for BLOCK0			
EFUSE_RD_WR_DIS_REG	BLOCK0 data register 0 that represents whether programming of individual eFuse memory bit is disabled	0x002C	RO
EFUSE_RD_REPEAT_DATA0_REG	BLOCK0 data register 1	0x0030	RO
EFUSE_RD_REPEAT_DATA1_REG	BLOCK0 data register 2	0x0034	RO
EFUSE_RD_REPEAT_DATA2_REG	BLOCK0 data register 3	0x0038	RO
EFUSE_RD_REPEAT_DATA3_REG	BLOCK0 data register 4	0x003C	RO
EFUSE_RD_REPEAT_DATA4_REG	BLOCK0 data register 5	0x0040	RO
Reading Data Registers for BLOCK1			
EFUSE_RD_MAC_SYS_0_REG	BLOCK1 data register 0	0x0044	RO
EFUSE_RD_MAC_SYS_1_REG	BLOCK1 data register 1	0x0048	RO
EFUSE_RD_MAC_SYS_2_REG	BLOCK1 data register 2	0x004C	RO
EFUSE_RD_MAC_SYS_3_REG	BLOCK1 data register 3	0x0050	RO
EFUSE_RD_MAC_SYS_4_REG	BLOCK1 data register 4	0x0054	RO
EFUSE_RD_MAC_SYS_5_REG	BLOCK1 data register 5	0x0058	RO
Reading Data Registers for BLOCK2			
EFUSE_RD_SYS_PART1_DATA0_REG	Register 0 for BLOCK2 (system)	0x005C	RO
EFUSE_RD_SYS_PART1_DATA1_REG	Register 1 for BLOCK2 (system)	0x0060	RO
EFUSE_RD_SYS_PART1_DATA2_REG	Register 2 for BLOCK2 (system)	0x0064	RO
EFUSE_RD_SYS_PART1_DATA3_REG	Register 3 for BLOCK2 (system)	0x0068	RO
EFUSE_RD_SYS_PART1_DATA4_REG	Register 4 for BLOCK2 (system)	0x006C	RO

Name	Description	Address	Access
EFUSE_RD_SYS_PART1_DATA5_REG	Register 5 for BLOCK2 (system)	0x0070	RO
EFUSE_RD_SYS_PART1_DATA6_REG	Register 6 for BLOCK2 (system)	0x0074	RO
EFUSE_RD_SYS_PART1_DATA7_REG	Register 7 for BLOCK2 (system)	0x0078	RO
Reading Data Registers for BLOCK3			
EFUSE_RD_USR_DATA0_REG	Register 0 for BLOCK3 (user)	0x007C	RO
EFUSE_RD_USR_DATA1_REG	Register 1 for BLOCK3 (user)	0x0080	RO
EFUSE_RD_USR_DATA2_REG	Register 2 for BLOCK3 (user)	0x0084	RO
EFUSE_RD_USR_DATA3_REG	Register 3 for BLOCK3 (user)	0x0088	RO
EFUSE_RD_USR_DATA4_REG	Register 4 for BLOCK3 (user)	0x008C	RO
EFUSE_RD_USR_DATA5_REG	Register 5 for BLOCK3 (user)	0x0090	RO
EFUSE_RD_USR_DATA6_REG	Register 6 for BLOCK3 (user)	0x0094	RO
EFUSE_RD_USR_DATA7_REG	Register 7 for BLOCK3 (user)	0x0098	RO
Reading Data Registers for BLOCK4			
EFUSE_RD_KEY0_DATA0_REG	Register 0 for BLOCK4 (KEY0)	0x009C	RO
EFUSE_RD_KEY0_DATA1_REG	Register 1 for BLOCK4 (KEY0)	0x00A0	RO
EFUSE_RD_KEY0_DATA2_REG	Register 2 for BLOCK4 (KEY0)	0x00A4	RO
EFUSE_RD_KEY0_DATA3_REG	Register 3 for BLOCK4 (KEY0)	0x00A8	RO
EFUSE_RD_KEY0_DATA4_REG	Register 4 for BLOCK4 (KEY0)	0x00AC	RO
EFUSE_RD_KEY0_DATA5_REG	Register 5 for BLOCK4 (KEY0)	0x00B0	RO
EFUSE_RD_KEY0_DATA6_REG	Register 6 for BLOCK4 (KEY0)	0x00B4	RO
EFUSE_RD_KEY0_DATA7_REG	Register 7 for BLOCK4 (KEY0)	0x00B8	RO
Reading Data Registers for BLOCK5			
EFUSE_RD_KEY1_DATA0_REG	Register 0 for BLOCK5 (KEY1)	0x00BC	RO
EFUSE_RD_KEY1_DATA1_REG	Register 1 for BLOCK5 (KEY1)	0x00C0	RO
EFUSE_RD_KEY1_DATA2_REG	Register 2 for BLOCK5 (KEY1)	0x00C4	RO
EFUSE_RD_KEY1_DATA3_REG	Register 3 for BLOCK5 (KEY1)	0x00C8	RO
EFUSE_RD_KEY1_DATA4_REG	Register 4 for BLOCK5 (KEY1)	0x00CC	RO
EFUSE_RD_KEY1_DATA5_REG	Register 5 for BLOCK5 (KEY1)	0x00D0	RO
EFUSE_RD_KEY1_DATA6_REG	Register 6 for BLOCK5 (KEY1)	0x00D4	RO
EFUSE_RD_KEY1_DATA7_REG	Register 7 for BLOCK5 (KEY1)	0x00D8	RO
Reading Data Registers for BLOCK6			
EFUSE_RD_KEY2_DATA0_REG	Register 0 for BLOCK6 (KEY2)	0x00DC	RO
EFUSE_RD_KEY2_DATA1_REG	Register 1 for BLOCK6 (KEY2)	0x00E0	RO
EFUSE_RD_KEY2_DATA2_REG	Register 2 for BLOCK6 (KEY2)	0x00E4	RO
EFUSE_RD_KEY2_DATA3_REG	Register 3 for BLOCK6 (KEY2)	0x00E8	RO
EFUSE_RD_KEY2_DATA4_REG	Register 4 for BLOCK6 (KEY2)	0x00EC	RO
EFUSE_RD_KEY2_DATA5_REG	Register 5 for BLOCK6 (KEY2)	0x00F0	RO
EFUSE_RD_KEY2_DATA6_REG	Register 6 for BLOCK6 (KEY2)	0x00F4	RO
EFUSE_RD_KEY2_DATA7_REG	Register 7 for BLOCK6 (KEY2)	0x00F8	RO
Reading Data Registers for BLOCK7			
EFUSE_RD_KEY3_DATA0_REG	Register 0 for BLOCK7 (KEY3)	0x00FC	RO
EFUSE_RD_KEY3_DATA1_REG	Register 1 for BLOCK7 (KEY3)	0x0100	RO
EFUSE_RD_KEY3_DATA2_REG	Register 2 for BLOCK7 (KEY3)	0x0104	RO

Name	Description	Address	Access
EFUSE_RD_KEY3_DATA3_REG	Register 3 for BLOCK7 (KEY3)	0x0108	RO
EFUSE_RD_KEY3_DATA4_REG	Register 4 for BLOCK7 (KEY3)	0x010C	RO
EFUSE_RD_KEY3_DATA5_REG	Register 5 for BLOCK7 (KEY3)	0x0110	RO
EFUSE_RD_KEY3_DATA6_REG	Register 6 for BLOCK7 (KEY3)	0x0114	RO
EFUSE_RD_KEY3_DATA7_REG	Register 7 for BLOCK7 (KEY3)	0x0118	RO
Reading Data Registers for BLOCK8			
EFUSE_RD_KEY4_DATA0_REG	Register 0 for BLOCK8 (KEY4)	0x011C	RO
EFUSE_RD_KEY4_DATA1_REG	Register 1 for BLOCK8 (KEY4)	0x0120	RO
EFUSE_RD_KEY4_DATA2_REG	Register 2 for BLOCK8 (KEY4)	0x0124	RO
EFUSE_RD_KEY4_DATA3_REG	Register 3 for BLOCK8 (KEY4)	0x0128	RO
EFUSE_RD_KEY4_DATA4_REG	Register 4 for BLOCK8 (KEY4)	0x012C	RO
EFUSE_RD_KEY4_DATA5_REG	Register 5 for BLOCK8 (KEY4)	0x0130	RO
EFUSE_RD_KEY4_DATA6_REG	Register 6 for BLOCK8 (KEY4)	0x0134	RO
EFUSE_RD_KEY4_DATA7_REG	Register 7 for BLOCK8 (KEY4)	0x0138	RO
Reading Data Registers for BLOCK9			
EFUSE_RD_KEY5_DATA0_REG	Register 0 for BLOCK9 (KEY5)	0x013C	RO
EFUSE_RD_KEY5_DATA1_REG	Register 1 for BLOCK9 (KEY5)	0x0140	RO
EFUSE_RD_KEY5_DATA2_REG	Register 2 for BLOCK9 (KEY5)	0x0144	RO
EFUSE_RD_KEY5_DATA3_REG	Register 3 for BLOCK9 (KEY5)	0x0148	RO
EFUSE_RD_KEY5_DATA4_REG	Register 4 for BLOCK9 (KEY5)	0x014C	RO
EFUSE_RD_KEY5_DATA5_REG	Register 5 for BLOCK9 (KEY5)	0x0150	RO
EFUSE_RD_KEY5_DATA6_REG	Register 6 for BLOCK9 (KEY5)	0x0154	RO
EFUSE_RD_KEY5_DATA7_REG	Register 7 for BLOCK9 (KEY5)	0x0158	RO
Reading Data Registers for BLOCK10			
EFUSE_RD_SYS_PART2_DATA0_REG	Register 0 for BLOCK10 (system)	0x015C	RO
EFUSE_RD_SYS_PART2_DATA1_REG	Register 1 for BLOCK10 (system)	0x0160	RO
EFUSE_RD_SYS_PART2_DATA2_REG	Register 2 for BLOCK10 (system)	0x0164	RO
EFUSE_RD_SYS_PART2_DATA3_REG	Register 3 for BLOCK10 (system)	0x0168	RO
EFUSE_RD_SYS_PART2_DATA4_REG	Register 4 for BLOCK10 (system)	0x016C	RO
EFUSE_RD_SYS_PART2_DATA5_REG	Register 5 for BLOCK10 (system)	0x0170	RO
EFUSE_RD_SYS_PART2_DATA6_REG	Register 6 for BLOCK10 (system)	0x0174	RO
EFUSE_RD_SYS_PART2_DATA7_REG	Register 7 for BLOCK10 (system)	0x0178	RO
Error Report Registers for BLOCK0			
EFUSE_RD_REPEAT_ERR0_REG	Programming error record register 0 for BLOCK0	0x017C	RO
EFUSE_RD_REPEAT_ERR1_REG	Programming error record register 1 for BLOCK0	0x0180	RO
EFUSE_RD_REPEAT_ERR2_REG	Programming error record register 2 for BLOCK0	0x0184	RO
EFUSE_RD_REPEAT_ERR3_REG	Programming error record register 3 for BLOCK0	0x0188	RO
EFUSE_RD_REPEAT_ERR4_REG	Programming error record register 4 for BLOCK0	0x018C	RO
Error Report Registers for RS Block			
EFUSE_RD_RS_ERR0_REG	Programming error record register 0 for BLOCK1-10	0x01C0	RO

Name	Description	Address	Access
EFUSE_RD_RS_ERR1_REG	Programming error record register 1 for BLOCK1-10	0x01C4	RO
eFuse Clock Register			
EFUSE_CLK_REG	eFuse clock configuration register	0x01C8	R/W
eFuse Configuration Registers			
EFUSE_CONF_REG	Configures eFuse operation mode	0x01CC	R/W
EFUSE_DAC_CONF_REG	Configures the eFuse programming voltage	0x01E8	R/W
EFUSE_RD_TIM_CONF_REG	Configures read timing parameters	0x01EC	R/W
EFUSE_WR_TIM_CONF1_REG	Configures eFuse programming timing parameters	0x01F0	R/W
EFUSE_WR_TIM_CONF2_REG	Configures eFuse programming timing parameters	0x01F4	R/W
EFUSE_WR_TIM_CONFO_RS_BYPASS_REG	Configures eFuse programming time parameters and RS bypass operation	0x01F8	varies
eFuse Status Register			
EFUSE_STATUS_REG	eFuse status register	0x01D0	RO
eFuse Command Registers			
EFUSE_CMD_REG	eFuse command register	0x01D4	varies
eFuse Interrupt Registers			
EFUSE_INT_RAW_REG	eFuse raw interrupt register	0x01D8	R/SS/WT
EFUSE_INT_ST_REG	eFuse interrupt status register	0x01DC	RO
EFUSE_INT_ENA_REG	eFuse interrupt enable register	0x01E0	R/W
EFUSE_INT_CLR_REG	eFuse interrupt clear register	0x01E4	WT
eFuse Version Register			
EFUSE_DATE_REG	eFuse version control register	0x01FC	R/W

8.6 Registers

The addresses in this section are relative to eFuse controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 8.1. EFUSE_PGM_DATA_{*n*}_REG (*n*: 0-7) (0x0000+0x4n*)**

<i>EFUSE_PGM_DATA_n</i>	
31	0
0x000000	
Reset	

EFUSE_PGM_DATA_{*n*} Configures the *n*th 32-bit data to be programmed. (R/W)

Register 8.2. EFUSE_PGM_CHECK_VALUE_{*m*}_REG (*m*: 0-2) (0x0020+0x4m*)**

<i>EFUSE_PGM_RS_DATA_m</i>	
31	0
0x000000	
Reset	

EFUSE_PGM_RS_DATA_{*m*} Configures the *m*th 32-bit RS code to be programmed. (R/W)

Register 8.3. EFUSE_RD_WR_DIS_REG (0x002C)

<i>EFUSE_WR_DIS</i>	
31	0
0x000000	
Reset	

EFUSE_WR_DIS Represents whether programming of individual eFuse memory bit is disabled.

- 1: Disabled
- 0: Enabled
- (RO)

Register 8.4. EFUSE_RD_REPEAT_DATA0_REG (0x0030)

[illegible]

EFUSE_RD_DIS Represents whether reading of individual eFuse block (BLOCK4 ~ BLOCK10) is disabled.

1: Disabled

0: Enabled

(RO)

EFUSE_USB_DEVICE_EXCHG_PINS Represents whether the USB Serial/JTAG pins D+ and D- are exchanged.

0: Not exchanged

1: Exchanged

(RO)

EFUSE_USB_OTG11_EXCHG_PINS Represents whether the USB FS pins D+ and D- are exchanged.

0: Not exchanged

1: Exchanged

(RO)

EFUSE_DIS_USB_JTAG Represents whether the USB-to-JTAG function in USB Serial/JTAG is disabled.

1: Disabled

0: Enabled

(RO)

EFUSE_POWERGLITCH_EN Represents whether power glitch detection is enabled.

0: Disabled

1: Enabled

(RO)

EFUSE_DIS_FORCE_DOWNLOAD Represents whether the function that forces chip into Download mode is disabled.

1: Disabled

0: Enabled

(RO)

EFUSE_SPI_DOWNLOAD_MSPI_DIS Represents accessing MSPI flash/MSPI RAM by SYS AXI matrix is disabled during boot_mode_download.

1: Disabled

0: Enabled

EFUSE_DIS_TWAI Represents whether TWAI function is disabled.

1: Disabled

Register 8.5. EFUSE_RD_REPEAT_DATA1_REG (0x0034)

EFUSE_KEY_PURPOSE_1																EFUSE_KEY_PURPOSE_0																EFUSE_SECURE_BOOT_KEY_REVOKE2																EFUSE_SECURE_BOOT_KEY_REVOKE1																EFUSE_SECURE_BOOT_KEY_REVOKE0																EFUSE_SPI_BOOT_CRYPT_CNT																EFUSE_WDT_DELAY_SEL																(reserved)																EFUSE_XTS_KEY_LENGTH_256																(reserved)															
31		28		27		24		23		22		21		20		18		17		16		15		14		13		0																																																																																																																																			
0x0				0x0				0		0		0		0x0				0x0				0		0		0		x										0		Reset																																																																																																																							

EFUSE_XTS_KEY_LENGTH_256 Represents which key is used for flash encryption.

0: XTS-256 key. Key length: 512 bits.

1: XTS-128 key. Key length: 256 bits.

(RO)

EFUSE_WDT_DELAY_SEL Represents RTC watchdog timeout threshold.

0: The originally configured STGO threshold × 2

1: The originally configured STGO threshold × 4

2: The originally configured STGO threshold × 8

3: The originally configured STGO threshold × 16

(RO)

EFUSE_SPI_BOOT_CRYPT_CNT Represents whether SPI boot encryption/decryption is enabled.

Odd count of bits with a value of 1: Enabled

Even count of bits with a value of 1: Disabled

(RO)

EFUSE_SECURE_BOOT_KEY_REVOKE0 Represents whether revoking Secure Boot key 0 is enabled.

1: Enabled

0: Disabled

(RO)

Continued on the next page...

Register 8.5. EFUSE_RD_REPEAT_DATA1_REG (0x0034)

Continued from the previous page...

EFUSE_SECURE_BOOT_KEY_REVOKE1 Represents whether revoking Secure Boot key 1 is enabled.

1: Enabled
0: Disabled
(RO)

EFUSE_SECURE_BOOT_KEY_REVOKE2 Represents whether revoking Secure Boot key 2 is enabled.

1: Enabled
0: Disabled
(RO)

EFUSE_KEY_PURPOSE_0 Represents the purpose of Key0. See Table 8.3-2. (RO)

EFUSE_KEY_PURPOSE_1 Represents the purpose of Key1. See Table 8.3-2. (RO)

Register 8.6. EFUSE_RD_REPEAT_DATA2_REG (0x0038)

EFUSE_FLASH_TPUW																						EFUSE_DIS_USB_OTG_DOWNLOAD_MODE										EFUSE_FLASH_ECC_EN										EFUSE_FLASH_PAGE_SIZE										EFUSE_FLASH_TYPE (reserved)										EFUSE_SECURE_BOOT_AGGRESSIVE_REVOKE										EFUSE_SECURE_BOOT_EN (reserved)										EFUSE_CRYPT_DPA_ENABLE										EFUSE_SEC_DPA_LEVEL										EFUSE_KEY_PURPOSE_5										EFUSE_KEY_PURPOSE_4										EFUSE_KEY_PURPOSE_3										EFUSE_KEY_PURPOSE_2									
31		28		27	26	25	24	23	22	21	20	19	18	17	16	15		12		11	8		7	4		3	0																																																																																																																		
0x0				0	0	0		0	0	0	0	0	1	0	0x0		0x0		0x0		0x0		0x0		0x0		0x0		Reset																																																																																																																

EFUSE_KEY_PURPOSE_2 Represents the purpose of Key2. See Table 8.3-2. (RO)

EFUSE_KEY_PURPOSE_3 Represents the purpose of Key3. See Table 8.3-2. (RO)

EFUSE_KEY_PURPOSE_4 Represents the purpose of Key4. See Table 8.3-2. (RO)

EFUSE_KEY_PURPOSE_5 Represents the purpose of Key5. See Table 8.3-2. (RO)

EFUSE_SEC_DPA_LEVEL Represents the security level of anti-DPA attack. The level is adjusted by configuring the clock random frequency division mode.

0: Security level is SEC_DPA_OFF

1: Security level is SEC_DPA_LOW

2: Security level is SEC_DPA_MIDDLE

3: Security level is SEC_DPA_HIGH

(RO)

EFUSE_CRYPT_DPA_ENABLE Represents whether defense against DPA attack is enabled.

1: Enabled

0: Disabled

(RO)

EFUSE_SECURE_BOOT_EN Represents whether Secure Boot is enabled.

1: Enabled

0: Disabled

(RO)

EFUSE_SECURE_BOOT_AGGRESSIVE_REVOKE Represents whether aggressive revocation of Secure Boot is enabled.

1: Enabled

0: Disabled

(RO)

Continued on the next page...

Register 8.6. EFUSE_RD_REPEAT_DATA2_REG (0x0038)

Continued from the previous page...

EFUSE_FLASH_TYPE Represents the type of the interfaced flash.

0: Four data lines

1: eight data lines

(RO)

EFUSE_FLASH_PAGE_SIZE Represents flash page size.

0: 256 bytes

1: 512 bytes

2: 1024 bytes

3: 2048 bytes

(RO)

EFUSE_FLASH_ECC_EN Represents whether ECC is enabled during flash boot.

0: Disabled

1: Enabled

(RO)

EFUSE_DIS_USB_OTG_DOWNLOAD_MODE Represents whether download via USB-OTG is disabled.

0: Enabled

1: Disabled

(RO)

EFUSE_FLASH_TPUW Represents the flash waiting time after power-up. Measurement unit: ms.

When the value is less than 15, the waiting time is the programmed value. Otherwise, the waiting time is a fixed value, i.e., 30 ms. (RO)

Register 8.7. EFUSE_RD_REPEAT_DATA3_REG (0x003C)

EFUSE_DDCD_VSET																EFUSE_HYS_EN_PAD																EFUSE_SECURE_BOOT_DISABLE_FAST_WAKE																EFUSE_SECURE_VERSION																EFUSE_FORCE_SEND_RESUME																EFUSE_UART_PRINT_CONTROL																EFUSE_ENABLE_SECURITY_DOWNLOAD																EFUSE_DIS_USB_SERIAL_JTAG_DOWNLOAD_MODE																EFUSE_DIS_DIRECT_BOOT																EFUSE_DIS_DOWNLOAD_MODE															
31				27				26		25		24																9				8		7		6		5		4		3		2		1		0																																																																																																															
0x0								0		0		0x00																0		0x0		0		0		0		0		0		0		0		0		Reset																																																																																																															

EFUSE_DIS_DOWNLOAD_MODE Represents whether all Download modes are disabled.

1: Disabled
0: Enabled
(RO)

EFUSE_DIS_DIRECT_BOOT Represents whether direct boot mode is disabled.

1: Disabled
0: Enabled
(RO)

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT Represents whether print from USB-Serial-JTAG during ROM boot is disabled.

1: Disabled
0: Enabled
(RO)

EFUSE_DIS_USB_SERIAL_JTAG_DOWNLOAD_MODE Represents whether the USB-Serial-JTAG download function is disabled.

1: Disabled
0: Enabled
(RO)

EFUSE_ENABLE_SECURITY_DOWNLOAD Represents whether security download is enabled. Only UART is supported for download. Reading/writing RAM or registers is not supported (i.e., stub download is not supported).

1: Enabled
0: Disabled
(RO)

Continued on the next page...

Register 8.7. EFUSE_RD_REPEAT_DATA3_REG (0x003C)

Continued from the previous page...

EFUSE_UART_PRINT_CONTROL Represents the type of UART printing.

- 0: Force enable printing.
 - 1: Enable printing when GPIO36 is reset at low level.
 - 2: Enable printing when GPIO36 is reset at high level.
 - 3: Force disable printing.
- (RO)

EFUSE_FORCE_SEND_RESUME Represents whether ROM code is forced to send a resume command during SPI boot.

- 1: Forced.
 - 0: Not forced.
- (RO)

EFUSE_SECURE_VERSION Represents the security version used by ESP-IDF anti-rollback feature.

(RO)

EFUSE_SECURE_BOOT_DISABLE_FAST_WAKE Represents whether FAST VERIFY ON WAKE is disabled when Secure Boot is enabled.

- 1: Disabled
 - 0: Enabled
- (RO)

EFUSE_HYS_EN_PAD Represents whether the hysteresis function of PADO – PAD27 is enabled.

- 1: Enabled
 - 0: Disabled
- (RO)

EFUSE_DCDC_VSET Represents the default DCDC voltage. (RO)

Register 8.8. EFUSE_RD_REPEAT_DATA4_REG (0x0040)

(reserved)										EFUSE_DIS_SWD EFUSE_DIS_WDT EFUSE_DCDC_VSET_EN EFUSE_HP_PWR_SRC_SEL										(reserved)										EFUSE_OPXA_TIEH_SEL_3 EFUSE_OPXA_TIEH_SEL_2 EFUSE_OPXA_TIEH_SEL_1 EFUSE_OPXA_TIEH_SEL_0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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EFUSE_OPXA_TIEH_SEL_0 Represents what controls the power supply for LDO V00.

- 0: The power supply is always high
 - 1: Controlled by the SDMMC1 peripheral
 - 2: Controlled by the PMU_LDO_VO_TIEH_0 register field
 - 3: Controlled by the SDMMCO peripheral
- (RO)

EFUSE_OPXA_TIEH_SEL_1 Represents what controls the power supply for LDO V01.

- 0: Controlled by the PMU_LDO_VO_TIEH_1 register field
 - 1: Controlled by the SDMMCO peripheral
 - 2: The power supply is always high
 - 3: Controlled by the SDMMC1 peripheral
- (RO)

EFUSE_OPXA_TIEH_SEL_2 Represents what controls the power supply for LDO V02.

- 0: Controlled by the PMU_LDO_VO_TIEH_2 register field
 - 1: Controlled by the SDMMCO peripheral
 - 2: The power supply is always high
 - 3: Controlled by the SDMMC1 peripheral
- (RO)

EFUSE_OPXA_TIEH_SEL_3 Represents what controls the power supply for LDO V03.

- 0: Controlled by the PMU_LDO_VO_TIEH_3 register field
 - 1: Controlled by the SDMMCO peripheral
 - 2: The power supply is always high
 - 3: Controlled by the SDMMC1 peripheral
- (RO)

Continued on the next page...

Register 8.8. EFUSE_RD_REPEAT_DATA4_REG (0x0040)

Continued from the previous page...

EFUSE_HP_PWR_SRC_SEL Represents the HP system power source.

0: LDO

1: DCDC

(RO)

EFUSE_DCDC_VSET_EN Represents whether to use the default voltage configured by [EFUSE_DCDC_VSET](#).

0: Not use

1: Use

(RO)

EFUSE_DIS_WDT Represents whether to disable the watchdog.

0: Enable

1: Disable

(RO)

EFUSE_DIS_SWD Represents whether to disable the super watchdog.

0: Enable

1: Disable

(RO)

Register 8.9. EFUSE_RD_MAC_SYS_O_REG (0x0044)**EFUSE_MAC_O** Represents the low 32 bits of MAC address. (RO)

EFUSE_MAC_EXT

EFUSE_MAC_1

EFUSE_MAC_EXT Represents the extended bits of MAC address. (RO)

EFUSE_MAC_RESERVED_0

EFUSE_MAC_RESERVED_1

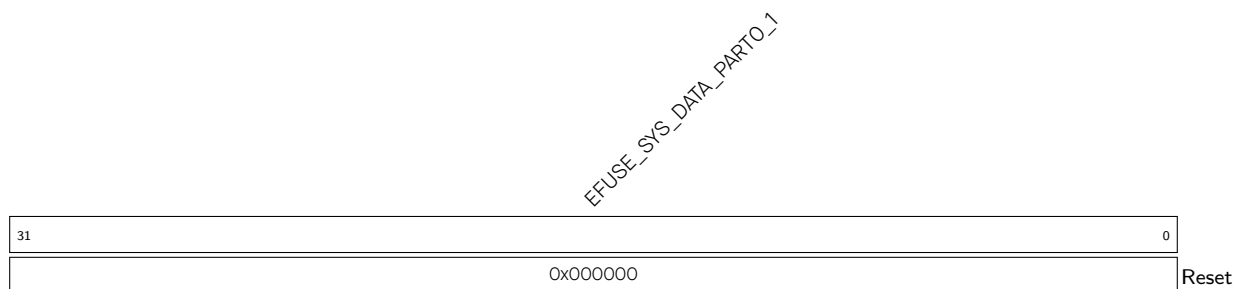
EFUSE_MAC_RESERVED_0 Reserved. (RO)

EFUSE_SYS_DATA_PART0_0

EFUSE_MAC_RESERVED_2

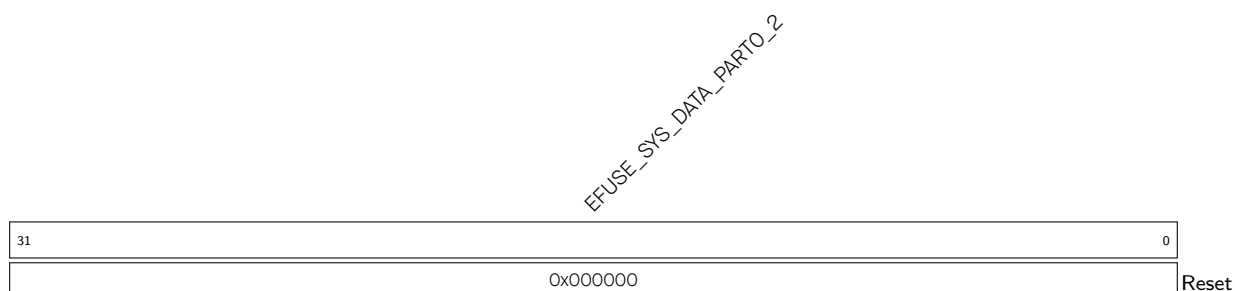
EFUSE_SYS_DATA_PART0_0 Represents the first 14 bits of the zeroth part of system data. (RO)

Register 8.13. EFUSE_RD_MAC_SYS_4_REG (0x0054)

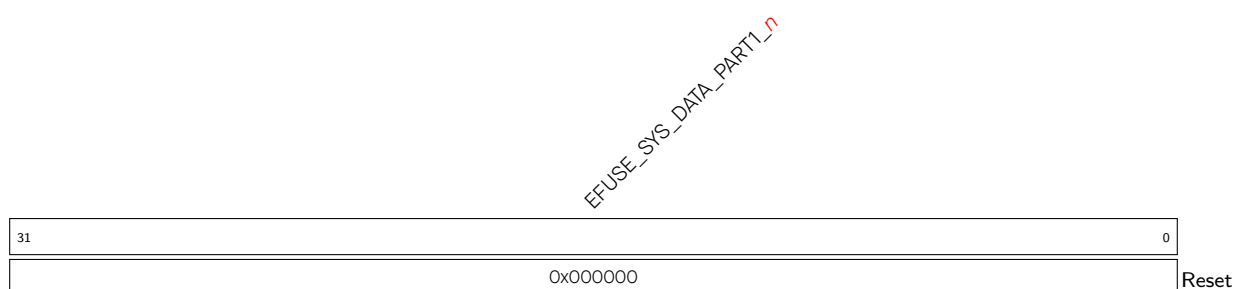


EFUSE_SYS_DATA_PART0_1 Represents the first 32 bits of the zeroth part of system data. (RO)

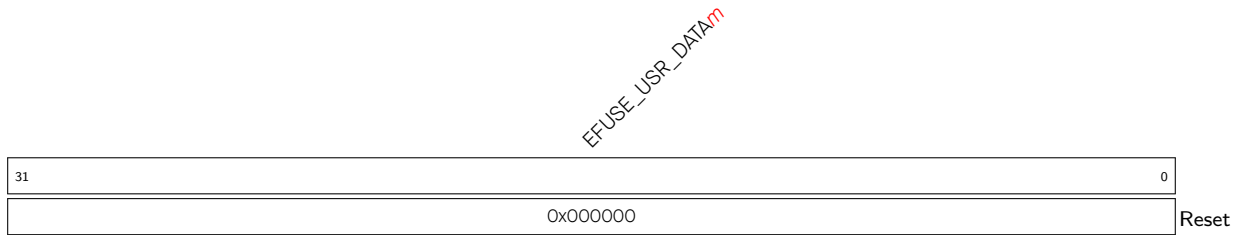
Register 8.14. EFUSE_RD_MAC_SYS_5_REG (0x0058)



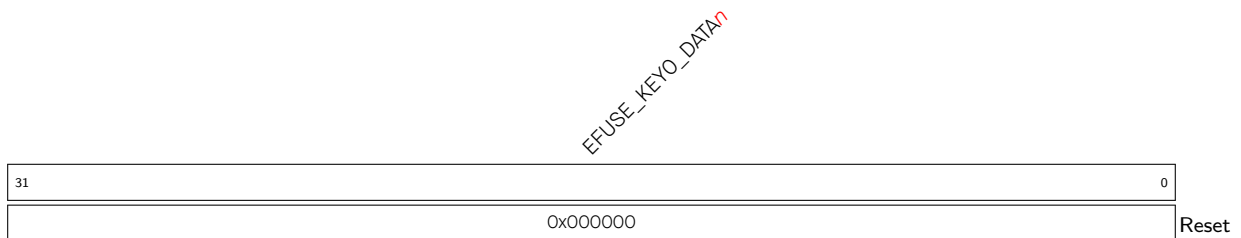
EFUSE_SYS_DATA_PART0_2 Represents the second 32 bits of the zeroth part of system data. (RO)

Register 8.15. EFUSE_RD_SYS_PART1_DATA_{*n*}_REG (*n*: 0-7) (0x005C+0x4**n*)

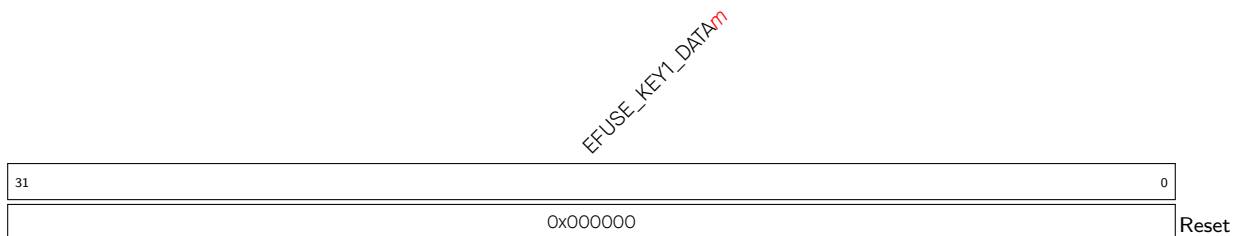
EFUSE_SYS_DATA_PART1_*n* Represents the *n*th 32 bits of the first part of system data. (RO)

Register 8.16. EFUSE_RD_USR_DATA m _REG (m : 0-7) (0x007C+0x4* m)

EFUSE_USR_DATA m Represents the m th 32 bits of BLOCK3 (user). (RO)

Register 8.17. EFUSE_RD_KEY0_DATA n _REG (n : 0-7) (0x009C+0x4* n)

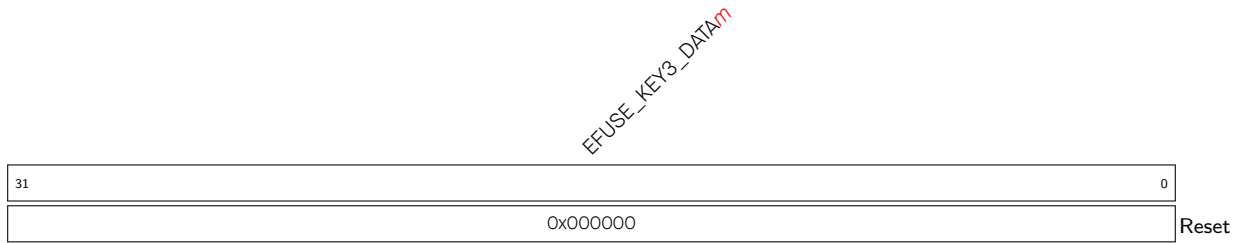
EFUSE_KEY0_DATA n Represents the n th 32 bits of KEY0. (RO)

Register 8.18. EFUSE_RD_KEY1_DATA m _REG (m : 0-7) (0x00BC+0x4* m)

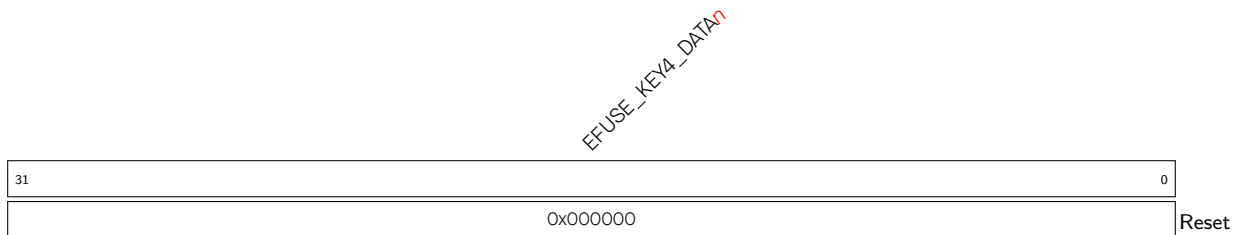
EFUSE_KEY1_DATA m Represents the m th 32 bits of KEY1. (RO)

Register 8.19. EFUSE_RD_KEY2_DATA n _REG (n : 0-7) (0x00DC+0x4* n)

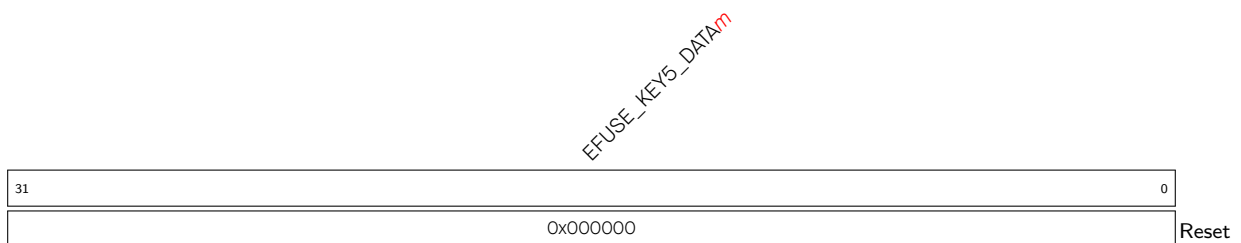
EFUSE_KEY2_DATA n Represents the n th 32 bits of KEY2. (RO)

Register 8.20. EFUSE_RD_KEY3_DATA m _REG (m : 0-7) (0x00FC+0x4* m)

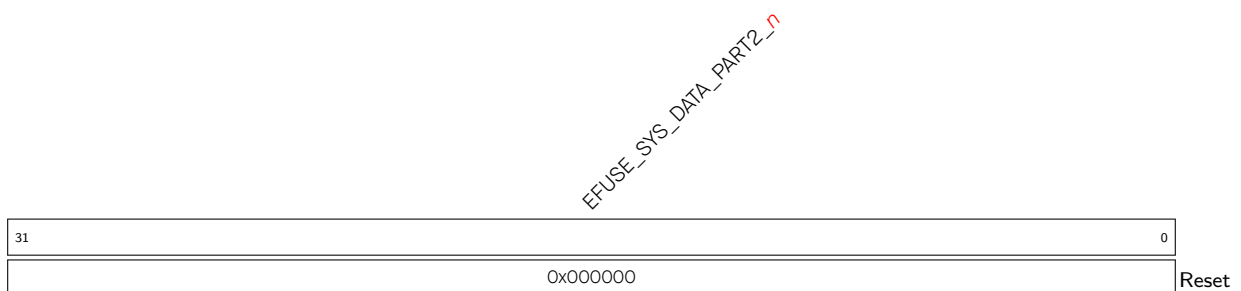
EFUSE_KEY3_DATA m Represents the m th 32 bits of KEY3. (RO)

Register 8.21. EFUSE_RD_KEY4_DATA n _REG (n : 0-7) (0x011C+0x4* n)

EFUSE_KEY4_DATA n Represents the n th 32 bits of KEY4. (RO)

Register 8.22. EFUSE_RD_KEY5_DATA m _REG (m : 0-7) (0x013C+0x4* m)

EFUSE_KEY5_DATA m Represents the m th 32 bits of KEY5. (RO)

Register 8.23. EFUSE_RD_SYS_PART2_DATA n _REG (n : 0-7) (0x015C+0x4* n)

EFUSE_SYS_DATA_PART2_ n Represents the n th 32 bits of the 2nd part of system data. (RO)

Register 8.24. EFUSE_RD_REPEAT_ERROR_REG (0x017C)

(reserved)																															EFUSE_USB_PHY_SEL_ERR				(reserved)			EFUSE_DIS_DOWNLOAD_MANUAL_ENCRYPT_ERR				EFUSE_DIS_PAD_JTAG_ERR			EFUSE_SOFT_DIS_JTAG_ERR			EFUSE_JTAG_SEL_ENABLE_ERR			EFUSE_DIS_TWAI_ERR			EFUSE_SPI_DOWNLOAD_ERR			(reserved)			EFUSE_FORCE_DOWNLOAD_ERR			EFUSE_POWERGLITCH_EN_ERR			EFUSE_DIS_USB_JTAG_ERR			EFUSE_DIS_USB_OTG11_EXCHG_PINS_ERR			EFUSE_DIS_USB_DEVICE_EXCHG_PINS_ERR			EFUSE_RD_DIS_ERR		
31						26	25	24	21			20	19	18	16			15	14	13	12	11	10	9	8	7	6	0																																																	
0x0						0	0	0	0	0	0	0	0	0x0			0	0	0	0	0	0	0	0	0	0	0	0x0						Reset																																											

EFUSE_RD_DIS_ERR Any bit of this field being 1 represents a programming error of RD_DIS. (RO)

EFUSE_DIS_USB_DEVICE_EXCHG_PINS_ERR This bit being 1 represents a programming error of DIS_USB_DEVICE_EXCHG_PINS. (RO)

EFUSE_DIS_USB_OTG11_EXCHG_PINS_ERR This bit being 1 represents a programming error of DIS_USB_OTG11_EXCHG_PINS. (RO)

EFUSE_DIS_USB_JTAG_ERR This bit being 1 represents a programming error of DIS_USB_JTAG.
(RO)

EFUSE_POWERGLITCH_EN_ERR This bit being 1 represents a programming error of POWER-GLITCH_EN. (RO)

EFUSE_DIS_FORCE_DOWNLOAD_ERR This bit being 1 represents a programming error of DIS_FORCE_DOWNLOAD. (RO)

EFUSE_SPI_DOWNLOAD_MSPI_DIS_ERR This bit being 1 represents a programming error of SPI_DOWNLOAD_MSPI_DIS. (RO)

EFUSE_DIS_TWAI_ERR This bit being 1 represents a programming error of DIS_TWAI. (RO)

EFUSE_JTAG_SEL_ENABLE_ERR This bit being 1 represents a programming error of JTAG_SEL_ENABLE. (RO)

EFUSE_SOFT_DIS_JTAG_ERR Any bit of this field being 1 represents a programming error of SOFT_DIS_JTAG. (RO)

EFUSE_DIS_PAD_JTAG_ERR This bit being 1 represents a programming error of DIS_PAD_JTAG.
(RO)

EFUSE_DIS_DOWNLOAD_MANUAL_ENCRYPT_ERR	This bit being 1 represents a programming error of DIS_DOWNLOAD_MANUAL_ENCRYPT. (RO)
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EFUSE_USB_PHY_SEL_ERR This bit being 1 represents a programming error of USB_PHY_SEL. (RO)

Register 8.25. EFUSE_RD_REPEAT_ERR1_REG (0x0180)

EFUSE_KEY_PURPOSE_1_ERR																
EFUSE_KEY_PURPOSE_0_ERR																
EFUSE_SECURE_BOOT_KEY_REVOKE2_ERR																
EFUSE_SECURE_BOOT_KEY_REVOKE1_ERR																
EFUSE_SECURE_BOOT_KEY_REVOKE0_ERR																
EFUSE_SPI_BOOT_CRYPT_CNT_ERR																
EFUSE_WDT_DELAY_SEL_ERR																
(reserved)																
EFUSE_XTS_KEY_LENGTH_256_ERR																
(reserved)																
31	28	27	24	23	22	21	20	18	17	16	15	14	13	0		
0x0		0x0		0	0	0	0x0	0x0	0	0	0x0					Reset

EFUSE_XTS_KEY_LENGTH_256_ERR This bit being 1 represents a programming error of XTS_KEY_LENGTH_256. (RO)

EFUSE_WDT_DELAY_SEL_ERR Any bit of this field being 1 represents a programming error of WDT_DELAY_SEL. (RO)

EFUSE_SPI_BOOT_CRYPT_CNT_ERR Any bit of this field being 1 represents a programming error of SPI_BOOT_CRYPT_CNT. (RO)

EFUSE_SECURE_BOOT_KEY_REVOKE0_ERR This bit being 1 represents a programming error of SECURE_BOOT_KEY_REVOKE0. (RO)

EFUSE_SECURE_BOOT_KEY_REVOKE1_ERR This bit being 1 represents a programming error of SECURE_BOOT_KEY_REVOKE1. (RO)

EFUSE_SECURE_BOOT_KEY_REVOKE2_ERR This bit being 1 represents a programming error of SECURE_BOOT_KEY_REVOKE2. (RO)

EFUSE_KEY_PURPOSE_0_ERR Any bit of this field being 1 represents a programming error of KEY_PURPOSE_0. (RO)

EFUSE_KEY_PURPOSE_1_ERR Any bit of this field being 1 represents a programming error of KEY_PURPOSE_1. (RO)

Register 8.26. EFUSE_RD_REPEAT_ERR2_REG (0x0184)

EFUSE_FLASH_TPUW_ERR																						EFUSE_DIS_USB_OTG_DOWNLOAD_MODE_ERR										EFUSE_FLASH_ECC_EN_ERR										EFUSE_FLASH_PAGE_SIZE_ERR										EFUSE_FLASH_TYPE_ERR (reserved)										EFUSE_SECURE_BOOT_AGGRESSIVE_REVOKE_ERR										EFUSE_SECURE_BOOT_EN_ERR										EFUSE_CRYPT_DPA_ENABLE_ERR										EFUSE_SEC_DPA_LEVEL_ERR										EFUSE_KEY_PURPOSE_5_ERR										EFUSE_KEY_PURPOSE_4_ERR										EFUSE_KEY_PURPOSE_3_ERR										EFUSE_KEY_PURPOSE_2_ERR									
31		28		27	26	25	24	23	22	21	20	19	18	17	16	15	12		11	8		7	4		3	0																																																																																																																			
0x0				0	0	0	0	0	0	0	0	0	0	0x0	0x0		0x0		0x0		0x0		0x0		0x0		Reset																																																																																																																		

EFUSE_KEY_PURPOSE_2_ERR Any bit of this field being 1 represents a programming error of KEY_PURPOSE_2. (RO)

EFUSE_KEY_PURPOSE_3_ERR Any bit of this field being 1 represents a programming error of KEY_PURPOSE_3. (RO)

EFUSE_KEY_PURPOSE_4_ERR Any bit of this field being 1 represents a programming error of KEY_PURPOSE_4. (RO)

EFUSE_KEY_PURPOSE_5_ERR Any bit of this field being 1 represents a programming error of KEY_PURPOSE_5. (RO)

EFUSE_SEC_DPA_LEVEL_ERR Any bit of this field being 1 represents a programming error of SEC_DPA_LEVEL. (RO)

EFUSE_CRYPT_DPA_ENABLE_ERR This bit being 1 represents a programming error of CRYPT_DPA_ENABLE. (RO)

EFUSE_SECURE_BOOT_EN_ERR This bit being 1 represents a programming error of SECURE_BOOT_EN. (RO)

EFUSE_SECURE_BOOT_AGGRESSIVE_REVOKE_ERR This bit being 1 represents a programming error of SECURE_BOOT_AGGRESSIVE_REVOKE. (RO)

EFUSE_FLASH_TYPE_ERR This bit being 1 represents a programming error of FLASH_TYPE. (RO)

EFUSE_FLASH_PAGE_SIZE_ERR This bit being 1 represents a programming error of FLASH_PAGE_SIZE. (RO)

EFUSE_FLASH_ECC_EN_ERR This bit being 1 represents a programming error of FLASH_ECC_EN. (RO)

EFUSE_DIS_USB_OTG_DOWNLOAD_MODE_ERR This bit being 1 represents a programming error of DIS_USB_OTG_DOWNLOAD_MODE. (RO)

EFUSE_FLASH_TPUW_ERR This bit being 1 represents a programming error of FLASH_TPUW. (RO)

Register 8.27. EFUSE_RD_REPEAT_ERR3_REG (0x0188)

31	27	26	25	24	9	8	7	6	5	4	3	2	1	0
0x0		0	0	0x00			0	0x0	0	0	0	0	0	0

EFUSE_DDC_VSET_ERR

EFUSE_HYS_EN_PAD_ERR

EFUSE_SECURE_BOOT_DISABLE_FAST_WAKE_ERR

EFUSE_SECURE_VERSION_ERR

EFUSE_FORCE_SEND_RESUME_ERR

EFUSE_UART_PRINT_CONTROL_ERR

EFUSE_ENABLE_SECURITY_DOWNLOAD_ERR

(reserved)

EFUSE_DIS_USB_SERIAL_JTAG_DOWNLOAD_MODE_ERR

EFUSE_DIS_DIRECT_BOOT_ERR

EFUSE_DIS_DOWNLOAD_MODE_ERR

Reset

EFUSE_DIS_DOWNLOAD_MODE_ERR This bit being 1 represents a programming error of DIS_DOWNLOAD_MODE. (RO)

EFUSE_DIS_DIRECT_BOOT_ERR This bit being 1 represents a programming error of DIS_DIRECT_BOOT. (RO)

EFUSE_DIS_USB_SERIAL_JTAG_ROM_PRINT_ERR This bit being 1 represents a programming error of DIS_USB_SERIAL_JTAG_ROM_PRINT_ERR. (RO)

EFUSE_DIS_USB_SERIAL_JTAG_DOWNLOAD_MODE_ERR This bit being 1 represents a programming error of DIS_USB_SERIAL_JTAG_DOWNLOAD_MODE. (RO)

EFUSE_ENABLE_SECURITY_DOWNLOAD_ERR This bit being 1 represents a programming error of ENABLE_SECURITY_DOWNLOAD. (RO)

EFUSE_UART_PRINT_CONTROL_ERR	Any bit of this field being 1 represents a programming error of UART_PRINT_CONTROL. (RO)
-------------------------------------	--

EFUSE_FORCE_SEND_RESUME_ERR This bit being 1 represents a programming error of FORCE_SEND_RESUME. (RO)

EFUSE_SECURE_VERSION_ERR Any bit of this field being 1 represents a programming error of SECURE VERSION. (RO)

EFUSE_SECURE_BOOT_DISABLE_FAST_WAKE_ERR	This bit being 1 represents a programming error of SECURE_BOOT_DISABLE_FAST_WAKE. (RO)
---	--

EFUSE_HYS_EN_PAD_ERR This bit being 1 represents a programming error of HYS_EN_PAD. (RO)

EFUSE_DCDC_VSET_ERR Any bit of this field being 1 represents a programming error of DCDC_VSET.
(RO)

Register 8.28. EFUSE_RD_REPEAT_ERR4_REG (0x018C)

(reserved)										EFUSE_DIS_SWD_ERR				EFUSE_DIS_WDT_ERR				EFUSE_DCDC_VSET_EN_ERR				EFUSE_HP_PWR_SRC_SEL_ERR				(reserved)				EFUSE_USB_OTG11_DREFL_ERR				EFUSE_USB_DEVICE_DREFL_ERR				(reserved)				EFUSE_OPXA_TIEH_SEL_3_ERR				EFUSE_OPXA_TIEH_SEL_2_ERR				EFUSE_OPXA_TIEH_SEL_1_ERR				EFUSE_OPXA_TIEH_SEL_0_ERR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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EFUSE_OPXA_TIEH_SEL_0_ERR Any bit of this field being 1 represents a programming error of OPXA_TIEH_SEL_0. (RO)

EFUSE_OPXA_TIEH_SEL_1_ERR Any bit of this field being 1 represents a programming error of OPXA_TIEH_SEL_1. (RO)

EFUSE_OPXA_TIEH_SEL_2_ERR Any bit of this field being 1 represents a programming error of OPXA_TIEH_SEL_2. (RO)

EFUSE_OPXA_TIEH_SEL_3_ERR Any bit of this field being 1 represents a programming error of OPXA_TIEH_SEL_3. (RO)

EFUSE_USB_DEVICE_DREFL_ERR Any bit of this field being 1 represents a programming error of USB_DEVICE_DREFL. (RO)

EFUSE_USB_OTG11_DREFL_ERR Any bit of this field being 1 represents a programming error of USB_OTG11_DREFL. (RO)

EFUSE_HP_PWR_SRC_SEL_ERR This bit being 1 represents a programming error of HP_PWR_SRC_SEL. (RO)

EFUSE_DCDC_VSET_EN_ERR This bit being 1 represents a programming error of DCDC_VSET_EN. (RO)

EFUSE_DIS_WDT_ERR This bit being 1 represents a programming error of DIS_WDT. (RO)

EFUSE_DIS_SWD_ERR This bit being 1 represents a programming error of DIS_SWD. (RO)

Register 8.29. EFUSE_RD_RS_ERR0_REG (0x01C0)

EFUSE_KEY4_FAIL		EFUSE_KEY4_ERR_NUM		EFUSE_KEY3_FAIL		EFUSE_KEY3_ERR_NUM		EFUSE_KEY2_FAIL		EFUSE_KEY2_ERR_NUM		EFUSE_KEY1_FAIL		EFUSE_KEY1_ERR_NUM		EFUSE_KEY0_FAIL		EFUSE_KEY0_ERR_NUM		EFUSE_USR_DATA_FAIL		EFUSE_USR_DATA_ERR_NUM		EFUSE_SYS_PART1_FAIL		EFUSE_SYS_PART1_ERR_NUM		EFUSE_MAC_SYS_FAIL		EFUSE_MAC_SYS_ERR_NUM	
31	30	28	27	26	24	23	22	20	19	18	16	15	14	12	11	10	8	7	6	4	3	2	0	Reset							
0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	0	0x0	Reset							

EFUSE_MAC_SYS_ERR_NUM Represents the number of error bytes. (RO)

EFUSE_MAC_SYS_FAIL Represents whether programming MAC_SYS failed.

0: No failure and the data of MAC_SYS is reliable.

1: Programming user data failed and the number of error bytes is over 6.

(RO)

EFUSE_SYS_PART1_ERR_NUM Represents the number of error bytes. (RO)

EFUSE_SYS_PART1_FAIL Represents whether programming system part1 data failed.

0: No failure and the data of system part1 is reliable.

1: Programming user data failed and the number of error bytes is over 6.

(RO)

EFUSE_USR_DATA_ERR_NUM Represents the number of error bytes. (RO)

EFUSE_USR_DATA_FAIL Represents whether programming user data failed.

0: No failure and the user data is reliable.

1: Programming user data failed and the number of error bytes is over 6.

(RO)

EFUSE_KEY0_ERR_NUM Represents the number of error bytes. (RO)

EFUSE_KEY0_FAIL Represents whether programming key0 data failed.

0: No failure and the data of key0 is reliable.

1: Programming key0 failed and the number of error bytes is over 6.

(RO)

EFUSE_KEY1_ERR_NUM Represents the number of error bytes. (RO)

EFUSE_KEY1_FAIL Represents whether programming key1 data failed.

0: No failure and the data of key1 is reliable.

1: Programming key1 failed and the number of error bytes is over 6.

(RO)

EFUSE_KEY2_ERR_NUM Represents the number of error bytes. (RO)

EFUSE_KEY2_FAIL Represents whether programming key2 data failed.

0: No failure and the data of key2 is reliable.

1: Programming key2 failed and the number of error bytes is over 6.

(RO)

Continued on the next page...

Register 8.29. EFUSE_RD_RS_ERR0_REG (0x01C0)

Continued from the previous page...

EFUSE_KEY3_ERR_NUM Represents the number of error bytes. (RO)**EFUSE_KEY3_FAIL** Represents whether programming key3 data failed.

0: No failure and the data of key3 is reliable.

1: Programming key3 failed and the number of error bytes is over 6.

(RO)

EFUSE_KEY4_ERR_NUM Represents the number of error bytes. (RO)**EFUSE_KEY4_FAIL** Represents whether programming key4 data failed.

0: No failure and the data of key4 is reliable.

1: Programming key4 failed and the number of error bytes is over 6.

(RO)

Register 8.30. EFUSE_RD_RS_ERR1_REG (0x01C4)

(reserved)																								EFUSE_SYS_PART2_FAIL				EFUSE_SYS_PART2_ERR_NUM				EFUSE_KEY5_FAIL				EFUSE_KEY5_ERR_NUM																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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EFUSE_KEY5_ERR_NUM Represents the number of error bytes. (RO)**EFUSE_KEY5_FAIL** Represents whether programming key5 data failed.

0: No failure and the data of key5 is reliable.

1: Programming key5 failed and the number of error bytes is over 6.

(RO)

EFUSE_SYS_PART2_ERR_NUM Represents the number of error bytes. (RO)**EFUSE_SYS_PART2_FAIL** Represents whether programming system part2 data failed.

0: No failure and the data of system part2 is reliable.

1: Programming user data failed and the number of error bytes is over 6.

(RO)

Register 8.31. EFUSE_CLK_REG (0x01C8)

[illegible]

EFUSE_MEM_FORCE_PD Configures whether to force power down eFuse SRAM.

1: Force

0: No effect

(R/W)

EFUSE_MEM_CLK_FORCE_ON Configures whether to force activate clock signal of eFuse SRAM.

1: Force activate

0: No effect

(R/W)

EFUSE_MEM_FORCE_PU Configures whether to force power up eFuse SRAM.

1: Force

0: No effect

(R/W)

EFUSE_CLK_EN Configures whether to force enable eFuse register configuration clock signal.

1: Force

0: The clock is enabled only during the reading and writing of registers

(R/W)

Register 8.32. EFUSE_CONF_REG (0x01CC)

31		20		19		16		15		0	
(reserved)				EFUSE_CFG_ECDSA_BLK				EFUSE_OP_CODE			
0 0 0 0 0 0 0 0 0 0 0 0				0x0				0x00			
Reset											

EFUSE_OP_CODE Configures operation command type.

0x5A5A: Program operation command

0x5AA5: Read operation command

Other values: No effect

(R/W)

EFUSE_CFG_ECDSA_BLK Configures which block to use for ECDSA key output. (R/W)

Register 8.33. EFUSE_DAC_CONF_REG (0x01E8)

(reserved)																EFUSE_OE_CLR								EFUSE_DAC_NUM								EFUSE_DAC_CLK_PAD_SEL								EFUSE_DAC_CLK_DIV							
31																	18	17	16									9	8	7																	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	255								0	0	7	23																Reset

EFUSE_DAC_CLK_DIV Configures the division factor of the rising clock of the programming voltage.
(R/W)

EFUSE_DAC_CLK_PAD_SEL Don't care. (R/W)

EFUSE_DAC_NUM Configures clock cycles for programming voltage to rise. Measurement unit: a clock cycle divided by [EFUSE_DAC_CLK_DIV](#). (R/W)

EFUSE_OE_CLR Configures whether to reduce the power supply of the programming voltage.
0: Not reduce.
1: Reduce.
(R/W)

Register 8.34. EFUSE_RD_TIM_CONF_REG (0x01EC)

EFUSE_READ_INIT_NUM																EFUSE_TSR_A								EFUSE_TRD								EFUSE_THR_A																																							
31																	24	23																	16	15																	8	7																	0
0xf																0x1																0x2																0x1																Reset							

EFUSE_THR_A Configures the read hold time. Measurement unit: One cycle of the eFuse core clock. (R/W)

EFUSE_TRD Configures the read time. Measurement unit: One cycle of the eFuse core clock.
(R/W)

EFUSE_TSR_A Configures the read setup time. Measurement unit: One cycle of the eFuse core clock. (R/W)

EFUSE_READ_INIT_NUM Configures the waiting time of reading eFuse memory. Measurement unit: One cycle of the eFuse core clock. (R/W)

Register 8.35. EFUSE_WR_TIM_CONF1_REG (0x01F0)

EFUSE_THP_A																EFUSE_PWR_ON_NUM																EFUSE_TSUP_A															
31								24								23								8								7								0							
0x1								0x2667																								0x1								Reset							

EFUSE_TSUP_A Configures the programming setup time. Measurement unit: One cycle of the eFuse core clock. (R/W)

EFUSE_PWR_ON_NUM Configures the power up time for VDDQ. Measurement unit: One cycle of the eFuse core clock. (R/W)

EFUSE_THP_A Configures the programming hold time. Measurement unit: One cycle of the eFuse core clock. (R/W)

Register 8.36. EFUSE_WR_TIM_CONF2_REG (0x01F4)

EFUSE_TPGM																EFUSE_PWR_OFF_NUM									
31															16									15	
0xa0																0x140									
Reset																									

EFUSE_PWR_OFF_NUM Configures the power outage time for VDDQ. Measurement unit: One cycle of the eFuse core clock. (R/W)

EFUSE_TPGM Configures the active programming time. Measurement unit: One cycle of the eFuse core clock. (R/W)

Register 8.37. EFUSE_WR_TIM_CONFO_RS_BYPASS_REG (0x01F8)

(reserved)												EFUSE_TPM_INACTIVE												EFUSE_UPDATE				EFUSE_BYPASS_RS_BLK_NUM												EFUSE_BYPASS_RS_CORRECTION																							
31												21												20												13				12				11												1				0			
0 0 0 0 0 0 0 0 0 0 0 0												0x1												0				0x0												0				Reset																			

EFUSE_BYPASS_RS_CORRECTION	Configures whether to bypass the Reed-Solomon (RS) correction step.
-----------------------------------	---

0: Not bypass

1: Bypass

(R/W)

EFUSE_BYPASS_RS_BLK_NUM	Configures which block number to bypass the Reed-Solomon (RS) correction step. (R/W)
--------------------------------	--

EFUSE_UPDATE Configures whether to update multi-bit register signals.

1: Update

0: No effect

(WT)

EFUSE_TPGM_INACTIVE Configures the inactive programming time. Measurement unit: One cycle of the eFuse core clock. (R/W)

Register 8.38. EFUSE_STATUS_REG (0x01D0)

Diagram illustrating the EFUSE register structure (32 bits):

- Bits 31-24: (reserved)
- Bits 23-19: EFUSE_CUR_ECDSA_BLK (Initial value: 0x0)
- Bits 10-9: EFUSE_BLK0_VALID_BIT_CNT (Initial value: 0x0)
- Bits 4-3: (reserved)
- Bits 0: EFUSE_STATE (Initial value: 0x0)

EFUSE_STATE Represents the state of the eFuse state machine.

0: Reset state, the initial state after power-up

1: Idle state

Other values: Non-idle state

(RO)

EFUSE_BLK0_VALID_BIT_CNT Represents the number of block valid bit. (RO)

EFUSE_CUR_ECDSA_BLK Represents which block is used for ECDSA key output. (RO)

Register 8.39. EFUSE_CMD_REG (0x01D4)

(reserved)																EFUSE_BLK_NUM		EFUSE_PGM_CMD		EFUSE_READ_CMD	
31															6	5	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0	0	0	Reset	

EFUSE_READ_CMD Configures whether to send read commands.

- 1: Send
 - 0: No effect
- (R/W/SC)

EFUSE_PGM_CMD Configures whether to send programming commands.

- 1: Send
 - 0: No effect
- (R/W/SC)

EFUSE_BLK_NUM Represents the serial number of the block to be programmed. Value 0-10 corresponds to block number 0-10, respectively. (R/W)

Register 8.40. EFUSE_INT_RAW_REG (0x01D8)

(reserved)																												EFUSE_PGM_DONE_INT_RAW		EFUSE_READ_DONE_INT_RAW																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
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EFUSE_READ_DONE_INT_RAW The raw interrupt status of EFUSE_READ_DONE_INT. (R/SS/WTC)

EFUSE_PGM_DONE_INT_RAW The raw interrupt status of EFUSE_PGM_DONE_INT. (R/SS/WTC)

Register 8.41. EFUSE_INT_ST_REG (0x01DC)

(reserved)																															EFUSE_PGM_DONE_INT_ST		EFUSE_READ_DONE_INT_ST																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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EFUSE_READ_DONE_INT_ST The masked interrupt status of EFUSE_READ_DONE_INT. (RO)

EFUSE_PGM_DONE_INT_ST The masked interrupt status of EFUSE_PGM_DONE_INT. (RO)

Register 8.42. EFUSE_INT_ENA_REG (0x01E0)

(reserved)																															EFUSE_PGM_DONE_INT_ENA		EFUSE_READ_DONE_INT_ENA																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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EFUSE_READ_DONE_INT_ENA Write 1 to enable EFUSE_READ_DONE_INT. (R/W)

EFUSE_PGM_DONE_INT_ENA Write 1 to enable EFUSE_PGM_DONE_INT. (R/W)

Register 8.43. EFUSE_INT_CLR_REG (0x01E4)

(reserved)																															EFUSE_PGM_DONE_INT_CLR		EFUSE_READ_DONE_INT_CLR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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EFUSE_READ_DONE_INT_CLR Write 1 to clear EFUSE_READ_DONE_INT. (WT)

EFUSE_PGM_DONE_INT_CLR Write 1 to clear EFUSE_PGM_DONE_INT. (WT)

Register 8.44. EFUSE_DATE_REG (0x01FC)

(reserved)				EFUSE_DATE																										
31				28	27																									0
0	0	0	0	0	0x2305050																									Reset

EFUSE_DATE Version control register. (R/W)